

Graph-Based Methods for Personalized News Recommendation Systems

A Final Project Mid Semester Report

submitted by

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DECLARATION OF ORIGINALITY

I, **Naga Satya Sai Sumakar Kothamasu**, with Roll No: **CS22B1077** hereby declare that the material presented in the Project Report titled **Graph-Based Methods for Personalized News Recommendation Systems** represents original work carried out by me in the **Department of Computer Science and Engineering** at the Indian Institute of Information Technology, Design and Manufacturing, Kancheepuram.

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Naga Satya Sai Sumakar Kothamasu

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ABSTRACT

The increase in growth of online news reading has led to a tremendous volume of information, and hence the need for a personalized news recommendation system that provides a better user experience. Traditional methods like collaborative filtering and content-based filtering have limitations in modeling complex user-item interaction and adapting to user interest dynamics. To address the limitations, recent developments in graph learning, particularly Graph Neural Networks (GNN), have shown promise in modeling complex user-item interaction and user interest dynamics. The goal of the project is to develop a personalized news recommendation system based on a graph learning paradigm.

The architecture for the project has been partially implemented based on a baseline architecture inspired by GREP, which captures existing and potential user interest. The architecture has shown promise in modeling user interaction and user interest dynamics, and its performance has been measured for personalized news recommendations. The performance has been measured using standard ranking metrics like MRR, nDCG@5, nDCG@10, AUC. The results have shown the promise of a graph learning paradigm for personalized news recommendations, and future improvements will be made to the architecture and its performance for a more robust and adaptable personalized news recommendation system.

KEYWORDS: Graph Neural Networks (GNN); Personalized Recommendation Systems; User Interest Modeling; Ranking Metrics.

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ABBREVIATIONS

ROC AUC	Receiver Operating Characteristic - Area Under Curve
BPR	Bayesian Personalized Ranking
CAUM	Candidate Aware User Modelling
CF	Collaborative Filtering
CNN	Convolution Neural Network
DivHGNN	Heterogeneous Graph Neural Network with Personalized and Adaptive Diversity
GNN	Graph Neural Network
GREP	Graph Neural News Recommendation with Existing and Potential Interest Modeling
MIND	Microsoft News Dataset
MF	Matrix Factorization
MRR	Mean Reciprocal Rank
NCF	Neural Collaborative Filtering
NRMS	Neural News Recommendation with Multi-Head Self-Attention
nDCG@5	Normalized Discounted Cumulative Gain at rank 5
nDCG@10	Normalized Discounted Cumulative Gain at rank 10
RNN	Recurrent Neural Network
SVD	Singular Value Decomposition

CHAPTER 1

Introduction

1.1 Problem Background

The sudden development of digital media platforms changed the way for the consumption of news. Online news portals provide thousands of articles every day on various domains such as politics, sports, finance, technology, entertainment, etc. Though the development of news articles has increased the accessibility, it has also led to the information overload problem. Users are not able to efficiently find the relevant and meaningful content without proper filtering.

The aim of a personalized news recommendation systems is addressing the information overload problem by modeling the user's preferences and providing the appropriate news articles. But, news recommendation faces many challenges that differentiate it from other recommendation systems such as product recommendation and movie recommendation:

- **Rapid Content Dynamics:** News articles have a short lifespan and they become outdated very quickly.
- **Evolving User Interests:** User preferences are dynamic and they can change over time.
- **Cold Start Problem:** Newly published articles doesn't have sufficient interaction history.
- **Rich Semantic Content:** News contains complex textual, categorical, and contextual information.
- **Need for Diversity:** Users expect a balanced mixture of relevant and diverse content.

Traditionally, collaborative filtering techniques are based on past user-item interaction data. Although these techniques are successful in some domains, they are not

suitable for news recommendations because of their inability to fully leverage semantic information contained in the news articles and their poor handling of dynamic user interests.

Recently, advancements in deep learning techniques are greatly benefiting content modeling techniques using attention models and neural networks. But most of these methods treat each user's news activity separately and cannot use the connections between users and news articles.

Graph learning techniques offer a promising direction for improvement because of their ability to leverage the relationships between users and news articles as nodes connected through a user-news interaction graph. By employing representation learning techniques, GNNs are able to leverage the relationships between users and news articles through their ability to model high-order connectivity.

The project is based on the recent developments in news recommendation techniques by applying various machine learning techniques. The main aim of this project is to utilize the relationships between the user and news articles by using graph learning techniques to improve the accuracy of news recommendations based on user interest.

1.2 Project Scope

The goal of my project is to build a personalized news recommendation system using a graph model and test its performance using different metrics of evaluation.

The project includes building a user-news interaction graph, designing a baseline recommendation model, and developing user representation with neighborhood aggregation and embedding propagation. The project also includes evaluating the performance of the model with respect to ranking measures, analyzing the performance of the model with existing models, and analyzing the modifications that can be made in the architecture for better performance.

This project focuses on research and experimentation rather than building a production ready system. It aims to study how GNNs can capture collaborative patterns and improve personalized news recommendations.

The goal is to better understand how graph-based representations work in dynamic news recommendation settings, while also considering the balance between model effectiveness and computational cost.

CHAPTER 2

Literature Survey

2.1 Introduction

Recommendation systems have become essential infrastructure for online platforms, whose function is to help mitigate the problem of information overload by presenting relevant information to individual users. It has seen tremendous progress from the initial heuristic and statistical approaches to the current GNN-based approaches that explicitly utilize the relational structure between users and items. In particular, the problem of news recommendation faces a number of unique challenges, including the high rate of content update, cold-start articles, and the changing nature of user interest. These challenges have given rise to a parallel line of research. This chapter reviews the underlying foundational techniques upon which this work builds. It starts with the traditional collaborative filtering techniques (Section 2.2), followed by the family of techniques based on deep learning (Section 2.3), GNN-based collaborative filtering (Section 2.4), and concludes with the three key models that this work draws upon: LightGCN, GREP, and DivHGNN (Sections 2.5, 2.6.2).

2.2 Traditional Collaborative Filtering

The first models of CF tried to utilize past user-item interactions to forecast future user preferences. User-based CF works by finding users who have similar interaction profiles and recommending items that those users have interacted with before. Item-based CF follows a similar paradigm to user-based CF but operates on the item level by recommending items similar to those a user has interacted with before. However, scalability issues are associated with this approach.

MF improved earlier methods by breaking the user item interaction matrix into low-dimensional representations for users and items. These latent vectors capture hidden

preferences, and recommendations are made by measuring their similarity. Later extensions like SVD++ further improved accuracy by incorporating users' past interactions. But, MF treats users and items as static and independent, without capturing relationships across multiple connections, led to development of graph, GNN based recommendations.

Wu et al. [1] identify the key shortcoming of MF as its inability to encode the collaborative signal latent in interaction patterns: two users who share many common interactions should have similar embeddings, yet nothing in the standard MF objective enforces this beyond their effect on the observed ratings. This observation directly motivates the shift towards graph based representation learning.

2.3 Deep Learning Based Recommendation

2.3.1 Neural Collaborative Filtering

Deep learning brought non linearity into the core of recommendation. NCF replaced the inner product of MF with a MLP to capture complex, non linear user item interactions, showed that learned interaction functions outperform linear counterparts on feedback benchmarks [1]. Convolutional and recurrent architectures were applied to side information such as item text and sequential click streams, further enriching latent representations.

Yet most neural CF models share a fundamental limitation with their MF predecessors: embeddings are derived only from the ID or attribute of the entity itself, without encoding the structure of interaction graph. Hence, two users who never co interact with the same item but are connected through chains of shared interactions remain unrelated in the learned embedding space.

2.3.2 Attention Based News Recommendation: NRMS

News recommendation demands a higher degree of content sensitivity than generic item recommendation, given articles have rich textual content, short lifecycles, and rapidly

evolving relevance. Wu et al. [2] suggested the **NRMS** framework, which applies Transformer style self attention at two levels.

At the word level, the MHSA mechanism identifies contextual dependencies between words in the news title, resulting in a context-aware word sequence. Additive attention then compresses the news title sequence into a news representation based on the importance weights of individual words. At the user level, the same mechanism is applied over the set of previously clicked articles to compress the users browsing history into a single user vector. The click probability for the candidate article is computed as the dot product between the candidate news embedding and the user embedding.

NRMS demonstrated consistent improvements over prior CNN and GRU based news encoders in AUC, MRR, nDCG@5, nDCG@10 on real world news clicks. Its simplicity and strong performance established it as a key baseline for subsequent graph enhanced news recommendation models.

2.3.3 Candidate Aware User Modelling: CAUM

Another limitation of methods like NRMS is user interest is modelled independently of the candidate news being evaluated. Qi et al. [3] says that users typically possess multiple simultaneous interests, and accurate recommendation requires matching the *specific* interest most relevant to a given candidate. Their CAUM approach injects the candidate news representation into the user encoder as a conditioning signal.

CAUM proposes three candidate aware components: (1) a candidate aware self attention network that uses the candidate article as a query to identify globally relevant clicked news; (2) a candidate aware CNN that models local behavioural context by incorporating candidate features into window level convolution; and (3) a candidate aware attention aggregator that weights previously clicked articles by their relevance to the candidate. This single tower architecture simultaneously models user and candidate during inference, achieving improved matching performance over two tower approaches. CAUM highlights the importance of query dependent interest aggregation in news recommendation.

2.3.4 Knowledge Graph Propagation: RippleNet

Structured external knowledge can enrich item representations beyond what click logs alone provide. Wang et al. [4] proposed **RippleNet**, which treats the KG as a source of preference propagation. Each user's seed preferences (directly interacted items) are expanded through iterative KG traversal: at hop h , a user's ripple set consists of all KG entities reachable within h steps from their clicked items. The relevance of entities in the ripple set to a candidate item is computed via a soft attention mechanism, and the accumulated evidence across all hops yields an enriched user representation. RippleNet showed the benefit of structured relational knowledge for improving recommendation quality, especially for users with sparse click histories.

2.4 Graph Neural Networks for Collaborative Filtering

2.4.1 Foundations

GNNs extend deep learning to graph structured data through a message passing paradigm: each node iteratively aggregates representations from its local neighbourhood and updates its own state accordingly. After K layers, a node's embedding encodes information from its K hop neighbourhood. Wu et al. [1] and Gao et al. [5] provide comprehensive taxonomies of GNN based recommenders, identifying four core design choices: graph construction, neighbour aggregation, information update, and final representation pooling.

In the recommendation context users, items are modelled as nodes and user item interaction is encoded as an edge. High order connectivity emerges naturally from GNN propagation: a user's embedding after two layers encodes not only their directly interacted items, but also items interacted with by users who share their interaction patterns precisely the collaborative filtering signal that MF was unable to capture properly.

2.4.2 NGCF

Wang et al. [6] proposed NGCF to directly include collaborative signals in the embedding learning process. NGCF builds a user item bipartite graph and spreads (propagates) embeddings on this graph. The update rule at layer l is:

$$\mathbf{e}_u^{(l)} = \sigma \left(\mathbf{W}_1 \mathbf{e}_u^{(l-1)} + \sum_{i \in \mathcal{N}_u} \frac{1}{\sqrt{|\mathcal{N}_u| |\mathcal{N}_i|}} \left(\mathbf{W}_1 \mathbf{e}_i^{(l-1)} + \mathbf{W}_2 (\mathbf{e}_i^{(l-1)} \odot \mathbf{e}_u^{(l-1)}) \right) \right) \quad (2.1)$$

where $\mathcal{N}_u$ means the items a user has interacted with u , $\mathbf{W}_1$, $\mathbf{W}_2$ denote trainable weight matrices, $\odot$ denotes element wise product, and σ is a non linear activation. The interaction term $\mathbf{e}_i^{(l-1)} \odot \mathbf{e}_u^{(l-1)}$ explicitly encodes first order feature interactions between users and items at each propagation step. Final embeddings are obtained by concatenating representations from all layers. NGCF demonstrated significant improvements over MF and attention based baselines on multiple public benchmarks.

2.5 LightGCN

NGCF showed that graph-based embedding propagation works well, but He et al. [7] said its design is too complex. Their experiments showed that feature transformation and non-linear activation do not improve results much and make training slower.

So, they proposed LightGCN, a simpler model. LightGCN removes weight matrices and activation functions and only keeps neighborhood aggregation. User and item embeddings are updated by averaging the embeddings of their connected neighbors in the graph.

It also uses normalization to stop high-degree nodes from dominating. Only the initial user and item embeddings are trained; all other layer embeddings are computed from them and the graph structure.

Layer Combination

Rather than using only the final layer’s embedding, LightGCN constructs the final representation as a weighted sum over all K layers:

$$\mathbf{e}_u = \sum_{k=0}^K \alpha_k \mathbf{e}_u^{(k)}, \quad \mathbf{e}_i = \sum_{k=0}^K \alpha_k \mathbf{e}_i^{(k)} \quad (2.2)$$

where $\alpha_k = 1/(K + 1)$ by default. This combination serves three purposes: it smoothly integrates neighbourhood signals at different hops, it contains the effect of self connections (avoiding the need for residual connections as an addition), and it removes oversmoothing by retaining a weighted contribution from the original embeddings. The final interaction score is the inner product $\hat{y}_{ui} = \mathbf{e}_u^\top \mathbf{e}_i$, optimised with BPR loss.

LightGCN achieved approximately 16% average relative improvement over NGCF across three public benchmarks under similar experimental conditions. The model’s linear structure also shows analytical insights: He et al. showed that the layer combination is equivalent to a polynomial filter on the normalised graph Laplacian, connecting LightGCN to spectral graph theory.

2.6 Graph Based News Recommendation

The graph based model has been extended specifically to news recommendation, where the interaction graph is augmented with entity, category, and semantic content. Two models: GREP and DivHGNN are of particular relevance to this work and are detailed below.

2.6.1 GREP: Existing and Potential Interest Modelling

Qiu et al. [8] proposed GREP, a graph-based news recommendation model, to solve a main problem of content-only news encoders. These models cannot use the connection structure between users and news in the interaction graph, especially for cold-start articles that have not received many clicks yet.

Architecture Overview

GREP constructs a knowledge graph $\mathcal{G}$ from entity co occurrence relationships in news articles and organises the recommendation pipeline into three major components.

The **title encoder** applies MHSA over word embeddings to produce a semantically rich news title representation. Entity embeddings derived from $\mathcal{G}$ via graph attention are then incorporated to produce the full news representation.

The **interest encoder** distinguishes between two types of user interest: *existing interests*, derived from the users click history, and *potential interests*, inferred by propagating user preferences through the entity knowledge graph to discover related but unobserved content. The interest encoder stacks L alternating layers: each layer updates the existing interest representation by attending over clicked news, and updates the potential interest representation by propagating through the KG neighbourhood of the users preferred entities. A bi directional interaction layer within each encoder block allows the two interest streams to mutually refine one another.

The **click predictor** calculates the score between the user’s combined interest representation and the candidate news embedding, with a negative sampling cross entropy objective.

Experimental Results

GREP was evaluated on the large scale MIND and Adressa datasets. It outperformed all baselines including NRMS, KRED, GNUM, and GERL on all metrics. Ablation studies confirmed that both the existing interest module and the potential interest module contribute independently to performance. Removing the bi directional interaction layer caused a notable drop, indicating that the mutual alignment between the two interest streams is important. Critically, GREP also outperformed graph based baselines GNUM, GERL that relied solely on the user news interaction graph without the KG, showing that entity level potential interest modelling is beneficial for new articles that have no interaction history.

2.6.2 DivHGNN: Heterogeneous Graph Modelling with Diversity

Zhang et al. [9] found two further limitations in prior graph based news recommendation models: they operate predominantly on homogeneous graphs (treating all node and edge types uniformly) and they ignore recommendation diversity, producing monotonous recommendation lists that fail to cover the breadth of a user's interests. **DivHGNN** addresses both limitations within a unified framework.

Attributed Heterogeneous Graph

DivHGNN constructs an attributed heterogeneous graph containing four distinct node types: users, news articles, words, and knowledge graph entities, connected by typed edges representing different relations (e.g., user news click, news entity mention, word news occurrence). Each node type carries heterogeneous attributes: textual attributes encoded via GloVe, knowledge attributes extracted from WikiData embeddings, and topological attributes produced by Node2Vec.

Heterogeneous Node Content Adapter (NCA)

Because the three attribute types exist in different embedding spaces and encode semantics at different levels of abstraction, directly aggregating them would introduce semantic gaps. The NCA addresses this by first projecting each attribute type through a type specific MLP aligner:

$$\tilde{a}_t^v = \text{MLP}_t(a_t^v) \quad (2.3)$$

Cross attribute dependencies are then modelled via self attention over the aligned attribute matrix $\tilde{C}^v$, and the result is fused by a shared MLP into a unified primitive node representation c_v . This design allows the subsequent GNN layers to operate on homogeneous dimensional inputs despite the underlying heterogeneity.

Heterogeneous Graph Representation Learning

After attribute alignment, DivHGNN performs typed neighbourhood aggregation. For each relation type, a relation specific attention network assigns importance weights to neighbour nodes. The aggregated neighbourhood signal is combined with the own representation of node through gating mechanism. Multiple relation specific aggregations are merged via a type level attention layer that learns the relative importance of each edge type for the target node.

Variational Representation and Diversity Modelling

To model the inherent uncertainty and diversity of user interests, DivHGNN employs variational representation learning: node embeddings are parameterised as Gaussian distributions $\mathcal{N}(\mu_v, \sigma_v^2)$ rather than point vectors. Sampling from these distributions during training regularises the representation space and encourages the model to encode a broader coverage of interests, which is then exploited at inference time through Personalised and Adaptive Diversity (PAD) modelling. PAD balances relevance and diversity in the final ranked list by incorporating individual level diversity objectives that penalise redundancy within a user’s recommendation list whilst still rewarding accuracy.

Experimental Results

DivHGNN was evaluated on the MIND dataset and demonstrated improvements over GREP, NRMS, and other graph based baselines in both AUC, nDCG and diversity metrics, showing that the heterogeneous graph architecture and variational diversity modelling complement each other.

2.7 Comparative Analysis and Research Gaps

The survey shows a clear trajectory: from matrix factorisation, to content aware deep models, to graph propagation methods that exploit interaction structure and finally het-

Table 2.1: Comparative summary of key recommendation models reviewed

Model	Graph Based	Heterogeneous	Content Aware	KG Enriched	Diversity
MF / SVD++	No	No	No	No	No
NRMS	No	No	Yes	No	No
CAUM	No	No	Yes	No	No
RippleNet	Yes	Yes	No	Yes	No
NGCF	Yes	No	No	No	No
LightGCN	Yes	No	No	No	No
GREP	Yes	Limited	Yes	Yes	No
DivHGNN	Yes	Yes	Yes	Yes	Yes

erogeneous, diversity aware frameworks. Yet, several research gaps remain. LightGCN and NGCF, while strong general purpose collaborative filters, lack any mechanism for incorporating the rich textual content of news articles. GREP, although integrating content and graph structure properly, operates on a homogeneous interaction graph augmented by a KG, and doesn't model recommendation diversity. DivHGNN's architectural complexity and computational overhead are considerable, raising practical concerns for large scale deployment. The temporal dynamics of news consumption users short term and long term interest shifts are not properly explored across all surveyed models.

The present work draws inspiration primarily from GREP's approach to combining content encoding with graph based interest propagation, and from LightGCN's principle that simplified propagation rules can outperform complex neural operations. The goal is to develop and evaluate a graph neural recommendation framework that strikes a tractable balance between structural expressiveness and practical efficiency, operating within a homogeneous graph setting suitable for direct comparison with established baselines on standard news recommendation benchmarks.

CHAPTER 3

Methodology

3.1 System Overview

The implementation is based on a graph-structured personalized news recommendation system following the GREP model. The system architecture includes the following major components:

- Data preprocessing and vocabulary building.
- News title encoding using multi-head self-attention.
- Construction of knowledge graph based on co-occurrence of entities.
- Graph-structured interest modeling using stacked Interest Encoding Blocks.
- Candidate scoring using gated fusion of semantic and structural representations.
- Candidate evaluation using ranking-based metrics.

The model represents users and news articles in a shared embedding space. User representation is constructed from historical clicked news, candidate news representation is generated using both textual and entity information. The final click probability is computed using dot product similarity between user and candidate representations.

3.2 Dataset Description (MIND Small)

Due to computational constraints, experiments were conducted on MIND-Small dataset, it's a subset of MIND-Large.

The dataset consists of:

- News articles with title, category, and entity annotations.
- User behavior logs containing clicked history and candidate impressions.

- Pre-trained entity embeddings.

For this implementation:

- Maximum title length: 20 words.
- Maximum user history length: 50 clicked articles.
- Negative sampling ratio: 4 negative samples per positive.

Entity co-occurrence in news articles was utilized to build a knowledge graph. Each entity node is connected to other entities in the same news article, limited to a maximum of 20 neighbors.

Table 3.1: MIND-Small Dataset

Metric	MIND-Small
Num. of users	50,000
Num. of news articles	51,282
# Entities	26,904
# Avg. Entities	3.03
# Categories	17
# Subcategories	264
# Avg. History Length	32.54
# Avg. Title Words	10.75
# Positive Samples	236,344
# Negative Samples	5,607,100

3.3 Implementation Details

The system is implemented in PyTorch and consists of modular components:

Title Encoder

The Title Encoder uses:

- Trainable word embedding layer (dimension = 100).
- Multi-head self-attention with 5 heads.
- Residual connection and layer normalization.
- Gated aggregation to produce a single news representation vector.

Interest Encoder

The Interest Encoder consists of stacked Interest Encoding Blocks, of which the number of layers is 2. Each block includes the following operations:

- Existing Interest Encoding by means of Multi-Head Attention.
- Potential Interest Encoding by means of a Graph Transformer Layer for the entity graph.
- Bidirectional interaction between titles and entities.

Prediction Mechanism

Final user representation is a combination of the following:

- Existing interest embedding.
- Potential interest embedding.

A gated fusion mechanism is applied before the computation of the dot product similarity with the candidate news representations.

3.4 Architecture 1: Lightweight GREP

The lightweight version of GREP reduces the complexity of the GREP architecture in the following ways:

- Fewer attention heads.
- Reduced complexity in graph propagation.
- Maintaining the embedding dimension at 100.
- Fewer interest layers.

This variant of GREP was created to minimize the computational cost without compromising the structural modeling.

Advantages:

- Faster training time.
- Lower memory consumption.

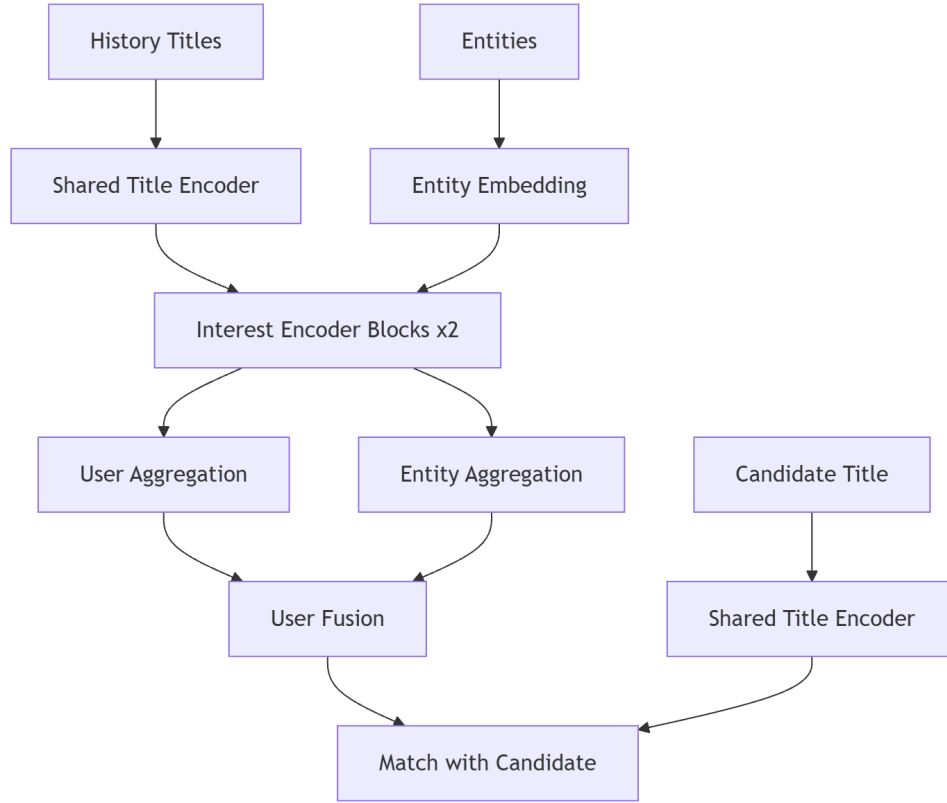


Figure 3.1: GREP Small

3.5 Architecture 2: Full GREP Implementation

The complete implementation is similar to the GREP architecture presented in the paper and includes:

- Multi-Head Self-Attention in title encoding.
- Two stacked Interest Encoding Blocks.
- Graph Transformer Layer for potential interest modeling.
- Bi-directional interaction between titles and entities.
- Gated fusion for user and news representations.

The training is done using the cross-entropy loss to approximate the InfoNCE loss with the positive candidate at index 0 among $k+1$ candidates. Additional stabilization techniques include:

- Gradient clipping with a maximum norm of 1.0.
- Replacement of NaN gradients with zero.
- Reduced learning rate scaling.

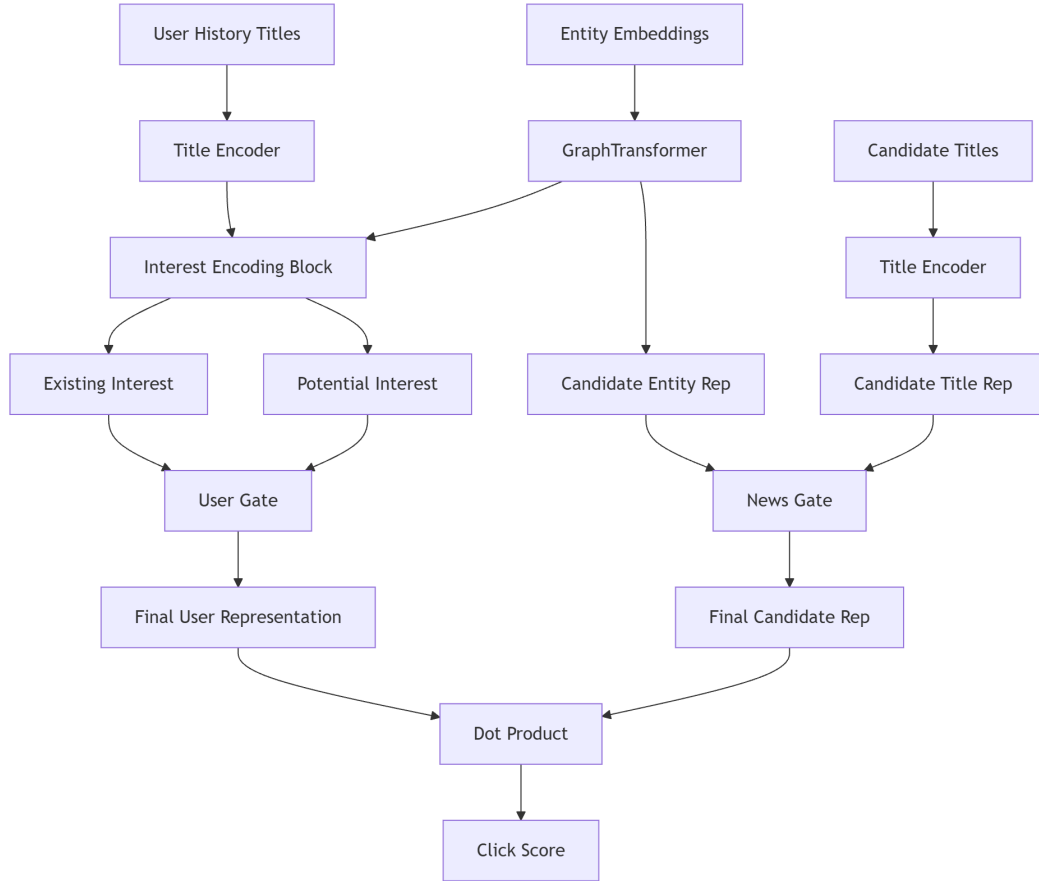


Figure 3.2: GREP Large

3.6 Experimental Setup

The setup used for the experiment is shown below:

- Embedding dimension: 100
- Title attention heads: 5
- Graph transformer heads: 2
- Interest layers: 2
- Batch size: 128
- Learning rate: 1×10^{-4}
- Adam Optimiser
- Dropout rate: 0.2
- Number of epochs: 5
- Device: CUDA

Evaluations used: AUC, MRR, nDCG@5, nDCG@10.

CHAPTER 4

Results, Analysis and Future Work

4.1 Results

Both implementations of the GREP architecture were trained and evaluated, and the metrics obtained are shown below (MIND Small dataset was used for both implementations)

Table 4.1: Result Comparison on MIND Small Dataset

Model	AUC	MRR	nDCG@5	nDCG@10
GREP Small	59.16	30.17	28.47	34.93
GREP Large	62.77	33.74	31.82	38.25
GREP (Paper, MIND Large)	69.44	34.40	37.54	43.22

It needs to be noted that the metrics obtained for the GREP model were based on the MIND Large dataset, which has a larger number of user interactions and news articles compared to the MIND Small dataset.

4.2 Observations

- The Full GREP is better than the Lightweight variant on all metrics.
- The performance gap between GREP Small and GREP Large supports the claim that structural modeling and stacked interest encoding blocks enhance the quality of the recommendations.
- The performance of the implemented GREP Large is close to the paper’s performance on the metrics (MRR: 33.74 vs 34.40), with the model trained on much smaller data than the paper (MIND Small instead of the original MIND dataset).
- The AUC and nDCG values still underperform the paper’s results, which is probably because the training data is smaller than the original.
- Structural modeling using entity-based graph propagation shows performance gains over the simple architecture.

4.3 Analysis

Thus, the performance improvement achieved by moving from GREP Small to GREP Large demonstrates the effectiveness of:

- Multi-layer Interest Encoding Blocks.
- Graph Transformer propagation over entity graph.
- Bi-directional interaction between titles and entities.
- Gated fusion of existing and potential interest representations.

Moreover, the achieved AUC of 62.77 is less than the AUC of 69.44 in the paper. This can be due to:

- Using MIND Small instead of MIND Large.
- Limited number of training epochs (5 epochs).
- Using random word embeddings instead of pre-trained word embeddings.
- Limited computational resources for hyperparameter tuning.

Interestingly, the obtained MRR value is very close to the paper's reported value. This indicates that the ranking quality for the top-click prediction task is good under the constraints of the dataset.

This demonstrates the effectiveness of the implemented architecture in capturing the semantic and structural information and validates the correctness of the implementation.

4.4 Current Limitations

4.4.1 Dataset:

- Only trained on MIND Small.
- Limited user interaction information, which limits the ability to generalize.
- Entity co-occurrence graph is limited due to the reduced size of the dataset.

4.4.2 Model:

- Word embeddings are randomly initialized instead of using pre-trained GloVe embeddings.
- Limited hyperparameter tuning.
- Limited interest encoding layers ($L=2$).

4.4.3 Experimental:

- Training was restricted to 5 epochs.
- No ablation study was conducted.

4.5 Future Plan

4.5.1 Dynamic and Temporal Interest Modeling

News consumption is a time-sensitive activity, and user interests can change over time. Future work will attempt to include temporal modeling mechanisms in the framework of the proposed graph model. This can be achieved by:

- The incorporation of recency-aware attention mechanisms.
- The application of temporal decay functions in the propagation of the graph.
- The modeling of the evolution of user interest using time-aware updates of the embeddings.

This aims to improve the modeling of short-term and evolving interests, which are critical in news recommendation systems.

4.5.2 Cold-Start Enhancement Mechanism

Although the sparsity of the graph is reduced to a certain extent, the cold-start problem is still a major challenge for new users and new news articles published in the near future. The areas that will be addressed in the near future include:

- Improving entity propagation for new news items that are introduced.

- Using knowledge-aware representations to improve sparse nodes.
- Evaluating the performance of the model in sparse user and sparse items cases.

This aims to improve robustness in low-interaction settings.

4.5.3 Structural Improvements Beyond Baseline GREP

The current design is based on the GREP approach; however, further architectural developments are planned:

- Investigate the possibility of adapting the graph propagation depth.
- Investigate the possibility of using different strategies for the fusion of the interest.
- Compare the performance of lightweight architectures with deeper architectures.

This will allow systematic analysis of how structural modifications influence ranking performance.

4.5.4 Representation Robustness and Contrastive Learning

To improve embedding quality and generalization, future work will explore:

- Integrating pre-trained word embeddings (GloVe) for better semantic encoding.
- Applying graph-based contrastive learning techniques.
- Studying robustness under noisy or partially observed interaction data.

These methods aim to enhance representation stability and ranking consistency.

4.5.5 Large-Scale and Cross-Dataset Evaluation

To validate scalability and generalization capability:

- Training and evaluation on the full MIND dataset.
- Performance benchmarking on additional public news datasets.
- Analyzing computational scalability and training stability.

4.6 Summary

Two GREP architectures have been successfully implemented in this project, and quantified benefits have been achieved through the use of graph-based modeling. GREP Large was able to obtain competitive performance in terms of MRR compared to the original paper, even though it was only trained on a smaller dataset.

Performance is expected to exceed the benchmark values as further optimization is done and scaling is applied to the MIND Large dataset.

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Weekly Review Report

Weekly Review Report

Roll No: CS22B1077
Name: Sai Sumakar

Week	Start Date - End Date	Work carried out during the week (with Your signature and Date)	Internal guide's comments with signature and Date
1	5/01/2026 - 11/01/2026	Problem Understanding, Literature Review - Recommendation systems <i>Sai</i> 10/1/26	
2	12/01/2026 - 18/01/2026	Review of Graph based Recommendation systems <i>Sai</i> 17/1/26	
3	19/01/2026 - 25/01/2026	Datasets Exploration and Understanding <i>Sai</i> 24/1/26	
4	26/01/2026 - 01/02/2026	Core modules Implementation (Title + Interest) <i>Sai</i> 31/1/26	
5	02/02/2026 - 08/02/2026	Experimenting with model design and hyperparameter config <i>Sai</i> 7/2/26	
6	9/02/2026 - 15/02/2026	Lightweight, & GREPLarge Implementations <i>Sai</i> 14/2/26	
7	16/02/2026 - 22/02/2026	Training & Evaluation <i>Sai</i> 21/2/26	
8	23/02/2026 - 01/03/2026	Results, Analysis, Report preparation and Future directions <i>Sai</i> 28/2/26	
9	02/03/2026 - 08/03/2026		
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